

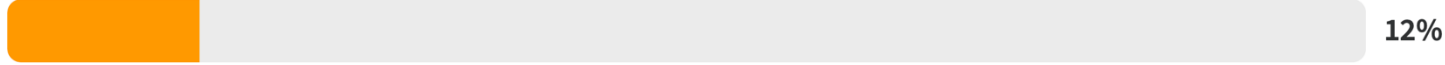
Lecture 4

Review Questions

An image patch contains closely spaced diagonal stripes. Which Gabor filter should generally produce the strongest response?



A low-frequency vertical filter



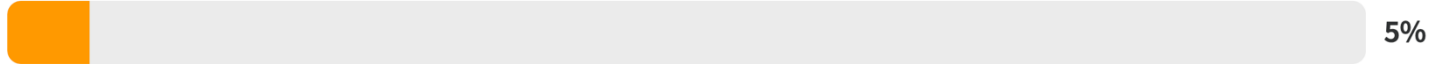
12%

A high-frequency diagonal filter



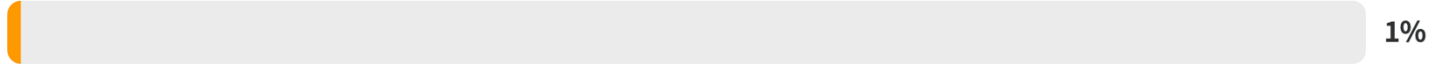
83%

A high-frequency horizontal filter



5%

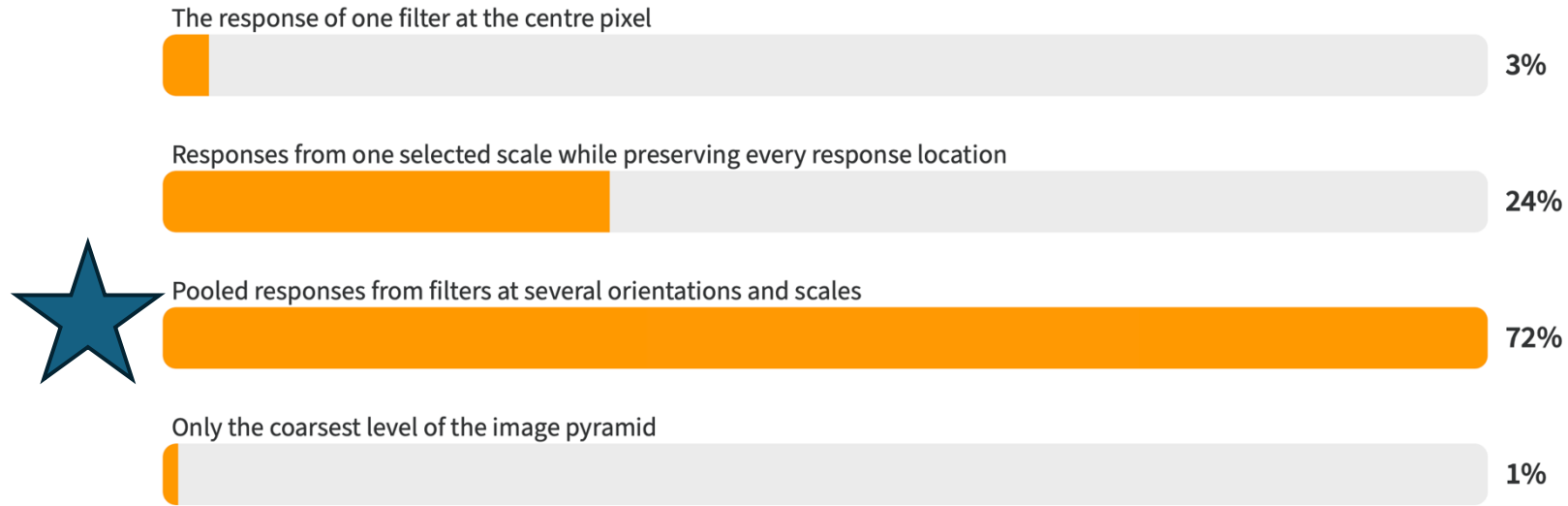
Any filter with a sufficiently large Gaussian envelope



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- The filter's orientation and spatial frequency should match the image structure. The Gaussian envelope controls spatial support, not the preferred stripe spacing.

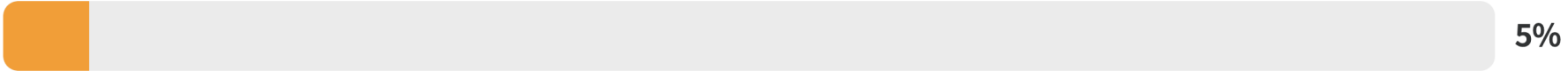
A fabric contains both fine fibres and a coarse weave, but their exact positions vary between patches. Which representation is most appropriate for material classification?



- Multiple scales capture both the fine and coarse structures. Pooling summarizes which patterns occur while reducing sensitivity to their exact positions.

Which statement correctly describes the roles of convolution, ReLU and pooling in a CNN?

Convolution performs pooling, while ReLU learns the kernels.



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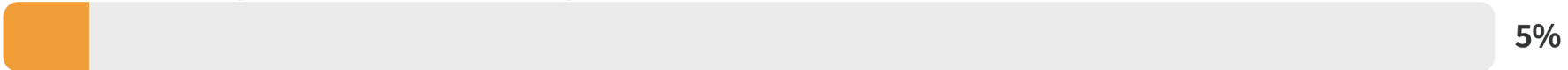


Convolution learns local filters, ReLU introduces nonlinearity, and pooling aggregates nearby responses.



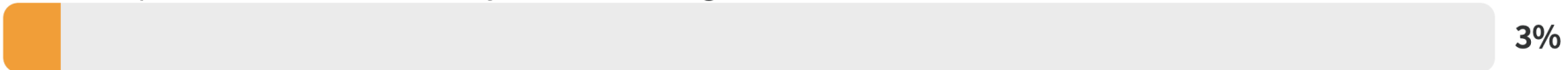
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ReLU reduces image resolution, while pooling learns new features



5%

All three operations are learned directly from the training data.



3%

- Convolutional kernels are learned. ReLU is a fixed nonlinear operation, and common pooling rules such as max pooling are also fixed; their architecture and parameters are selected by the designer.